

House Price Prediction - Project Summary

Objective:

The objective of this project was to build a machine learning model capable of predicting house prices using property-related features such as area, bedrooms, bathrooms, stories, parking availability, furnishing status, and amenities.

Data Preparation:

The housing dataset containing 545 records was loaded and explored using Pandas. Missing values and duplicate records were checked, and categorical variables were converted into numerical form using one-hot encoding.

Model Development:

The dataset was split into training and testing sets using an 80:20 ratio. Two regression models were trained and evaluated: Linear Regression and Random Forest Regressor.

Results:

- Linear Regression: MAE = 970,043, RMSE = 1,324,507, $R^2 = 0.653$
- Random Forest: MAE = 1,021,546, RMSE = 1,400,566, $R^2 = 0.612$

Linear Regression delivered the best performance and was selected as the final model.

Key Findings:

- Area was the most influential factor affecting house prices.
- Bathrooms, air conditioning, parking availability, and stories also contributed significantly.
- Most properties were concentrated in the lower to mid-price range, with a few high-priced outliers.

Recommendation:

Real estate businesses should focus on property size and important amenities when estimating market value and developing pricing strategies.